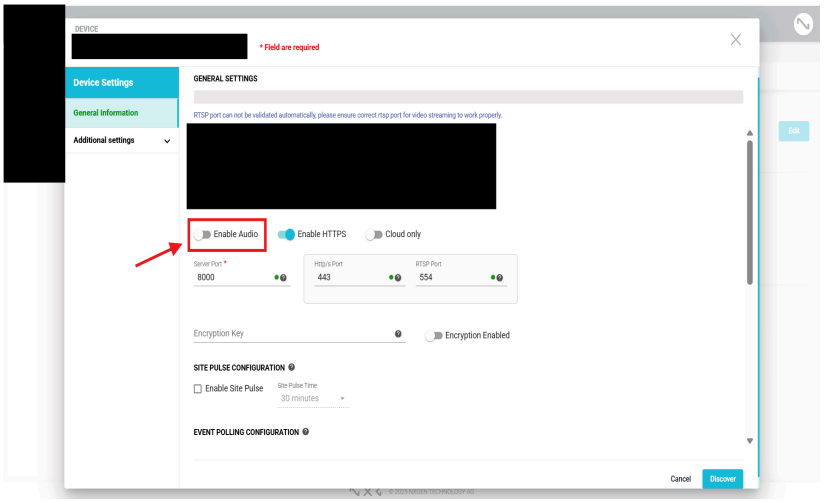
Hikvision audio Implementation

Configuration:

In Genesis, audio transmission can be configured at two levels — Device level and Sensor level— each determining which type of speaker channels are used.

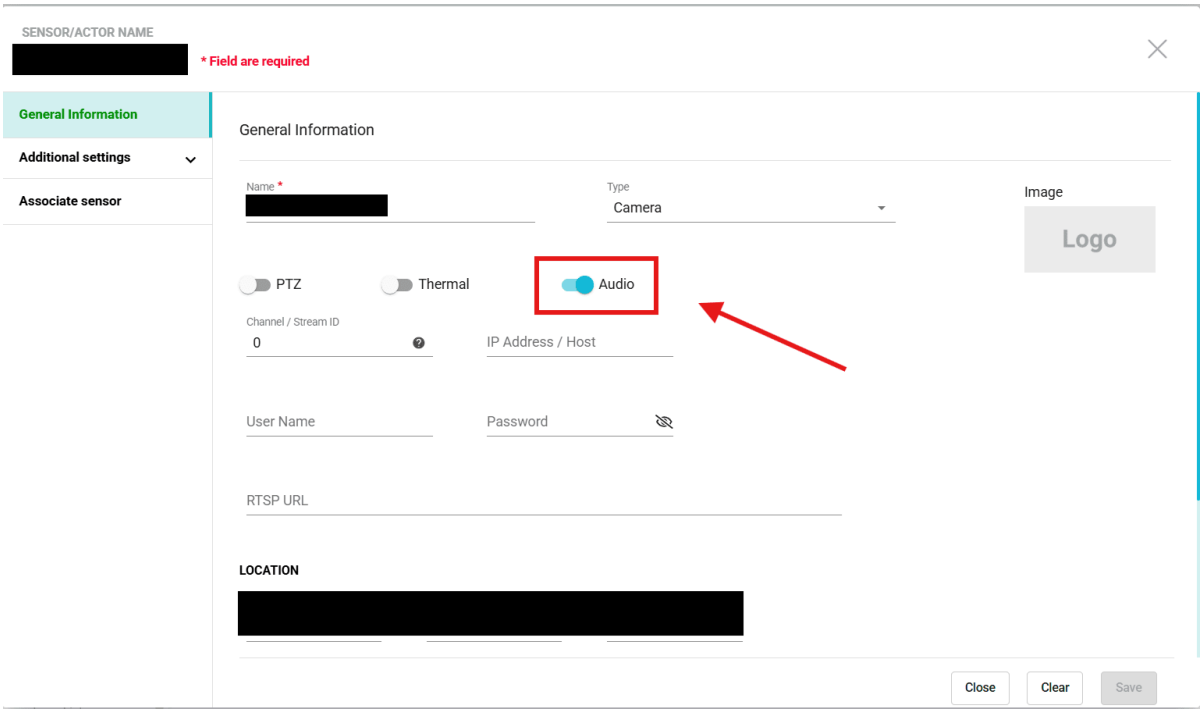
1. Device-level Configuration:

When audio is enabled at the device level, Genesis uses speaker which is connected directly to the device and are common to all cameras on that device for audio transmission.
Discover and save while enabling the audio toggle. *(Refer to the image below for an example of the device-level configuration.)*



2. Sensor-level configuration:

When audio is enabled at the sensor (camera) level, Genesis uses speakers which are linked specifically to individual cameras and can only be used with those cameras.
Always save after enabling the audio toggle.
(Refer to the image below for an example of the sensor-level configuration.)



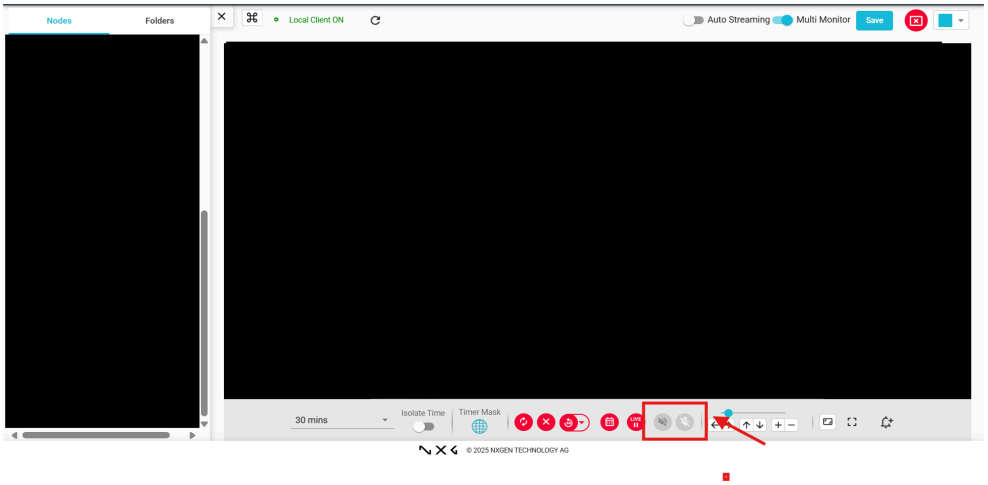
3. Both Configurations Enabled:

If both device-level and sensor-level audio are enabled, Genesis will display all available speaker channels for the selected camera in the Video Viewer page.

4. Both Configuration Disabled:

If both **device-level** and **sensor-level** audio are disabled:

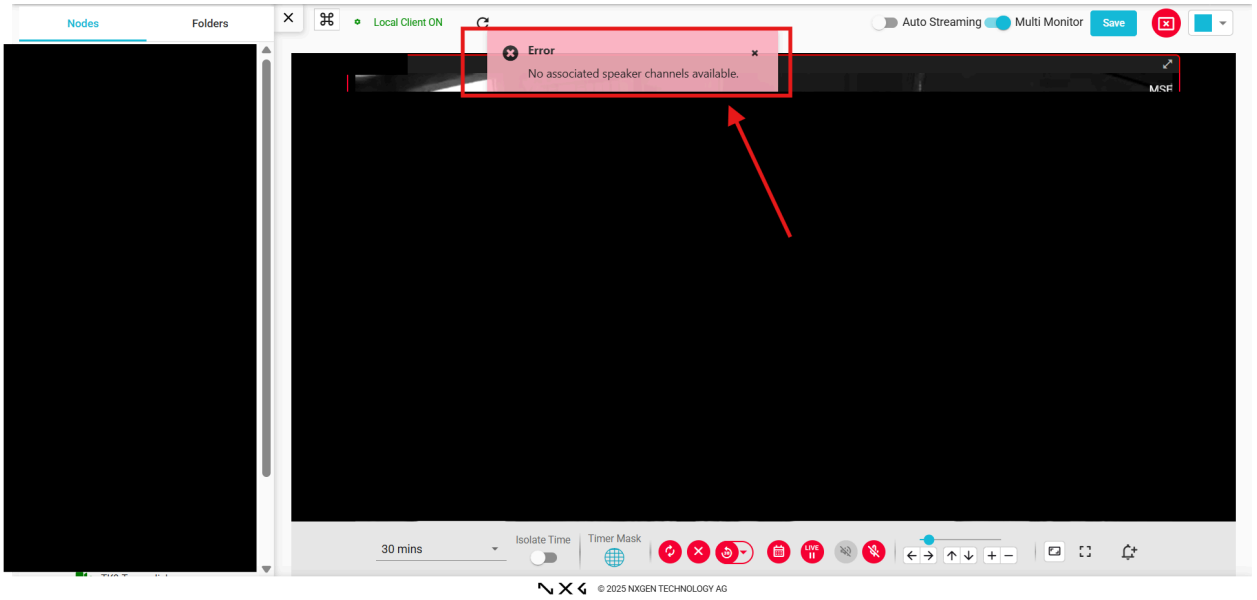
- The Mic button in the Video Viewer will be disabled.
- Users will not be able to select any speaker channel or turn on the microphone.
- To enable audio transmission, at least one of the configurations must be enabled. *(Refer to the image below for reference.)*



5. Sensor-level configuration (Device level Disabled):

When **sensor-level audio** is enabled but **device-level audio** is disabled, the following applies:

- If the camera has a linked speaker, the microphone will operate using that specific speaker channel.
- If no speaker is linked to the camera:
  - The Mic button in the Video Viewer will still appear enabled.
  - When the user attempts to turn on the microphone, an error message will appear:
  - *"No associated speaker channels available". (Refer to the image below for reference.)*

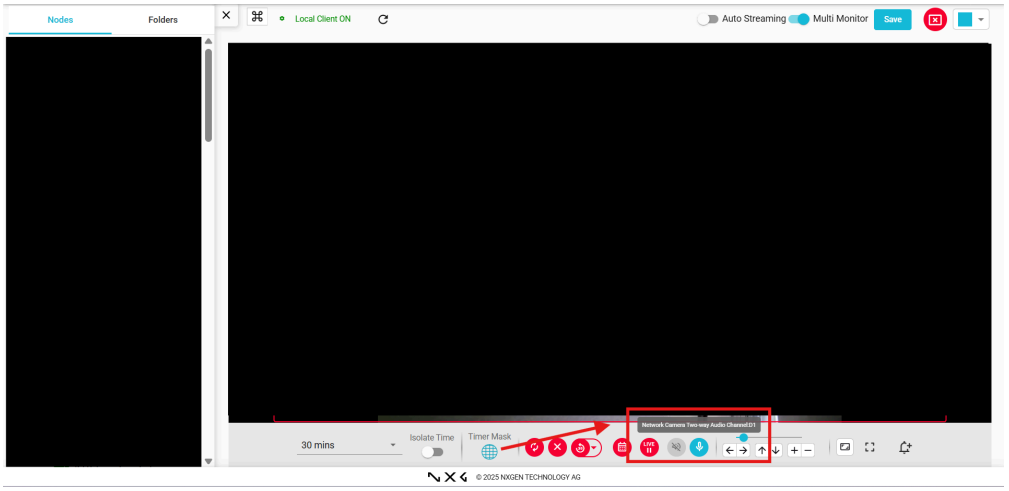


Video Viewer:

In the Video Viewer, the available speaker channels are displayed based on the active audio configuration (Device level, Sensor level, or both).

Channel Selection

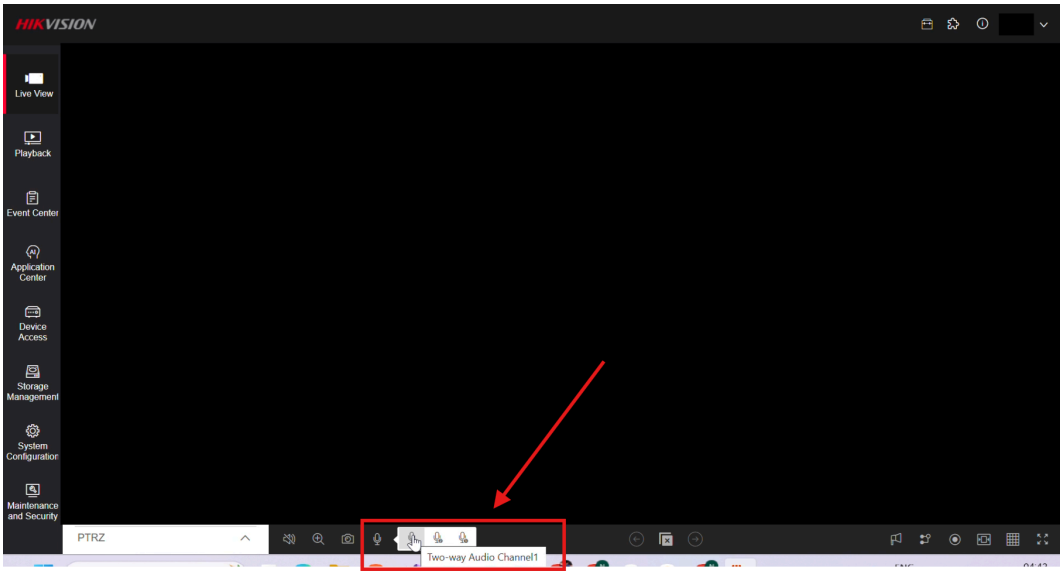
- Users can select a specific speaker channel and activate the microphone by clicking the Mic button.
- Hovering over a speaker icon shows the speaker name.
- If the speaker is active, the tooltip also shows the enabled speaker channel name.
- Clicking the Mic button again will turn off the audio. *(Refer to the image below for the visual representation.)*

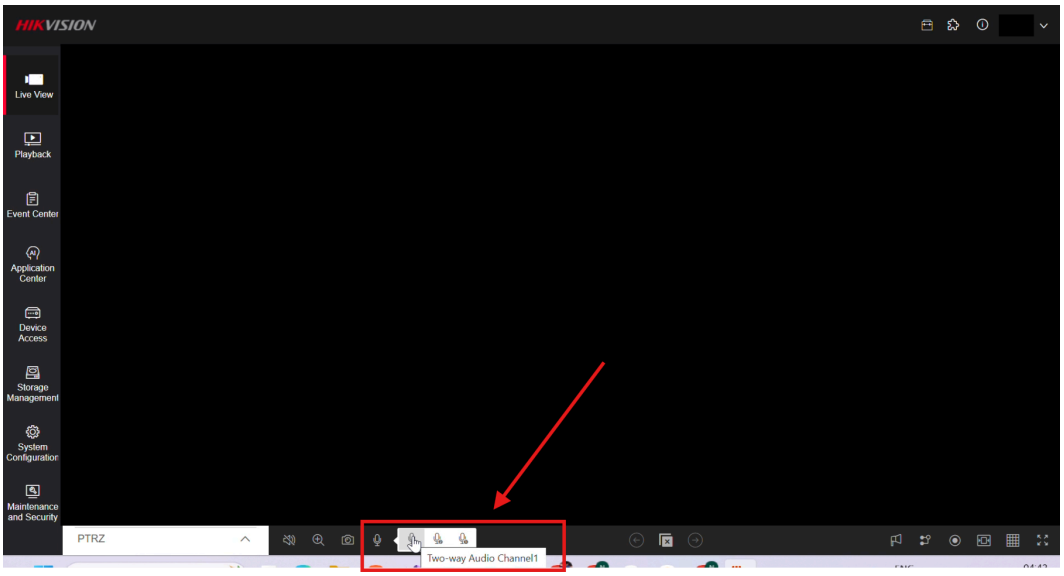


Web Client:

- When logging into a device in the web client, the functionality provided for enabling audio transmission in the Video Viewer is the same as in Genesis.
- If the device has only a device-level speaker, the UI will display the corresponding available speaker.
- If the device has a speaker linked to the camera, the UI will display all speakers available for that specific camera.
- Speaker names are matched exactly as they appear in the web client, ensuring consistency between the platforms. *(Refer to the image below for reference.)*

Note: The attached screenshot was taken from the Hikvision web client running firmware version V5.04.066 build 250620.





Important note:

If a site has **Genesis audio** enabled, all devices under that site will use Genesis audio as the **first priority**.

- This overrides both **device-level** and **sensor-level** speakers.
- Audio can be transmitted directly without selecting any channel.
- If Genesis audio is **not available** or is **disabled**, the system will fall back to using the device-level and sensor-level audio configurations.